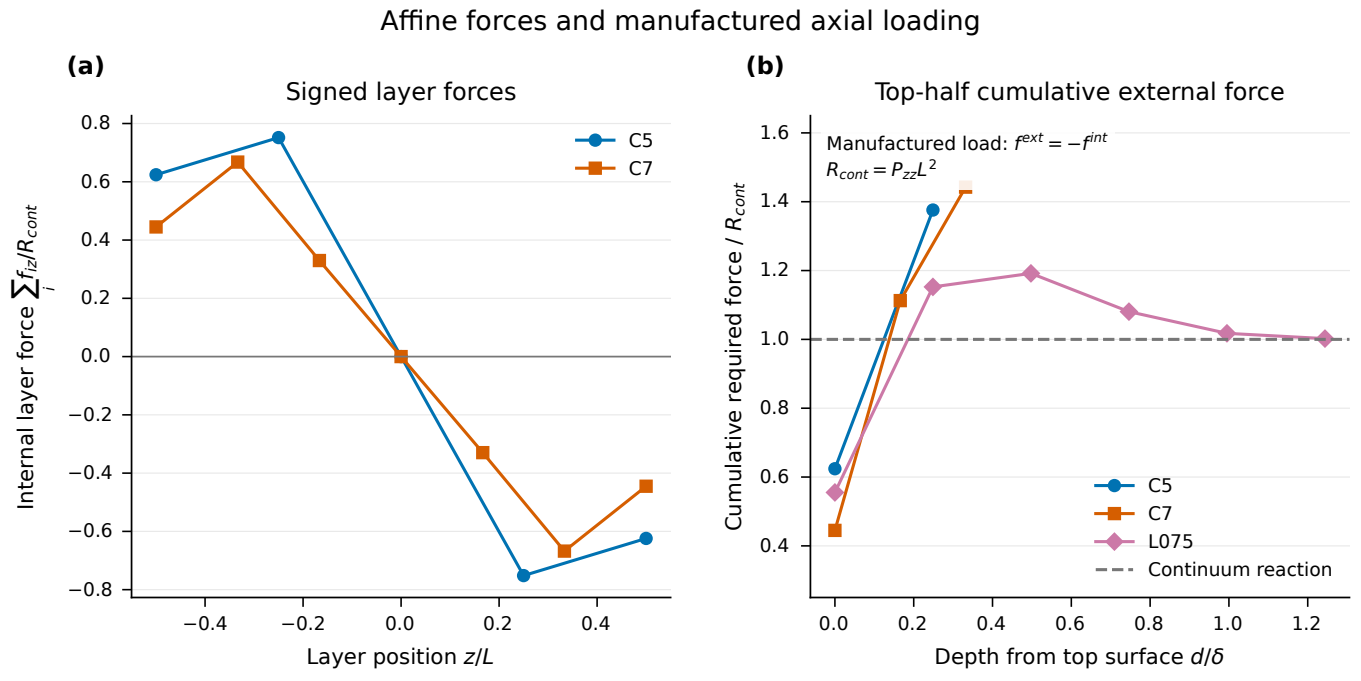
Affine forces and manufactured axial loading
(a)
Signed layer forces
C5
C7
Internal layer force $\sum_i f_{iz}/R_{cont}$
Layer position z/L
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−0.4
−0.2
0.0
0.2
0.4
(b)
Top-half cumulative external force
Manufactured load: $f^{ext} = -f^{int}$
$R_{cont} = P_{zz}L^2$
Cumulative required force / R_{cont}
Depth from top surface d/δ
C5
C7
L075
Continuum reaction
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.0
0.2
0.4
0.6
0.8
1.0
1.2